



BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)

FINAL EXAMINATION JUNE 2025

Course : EC3114 (Software Modelling and Analysis) Time: 8.00 am – 11.00 am
(3 hours)
Lecturer : Mr. Izrulfizal Saufihamizal bin Ibrahim Date : 17 November 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 4 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1.
 - a. Explain the benefits of using software modeling in a software development project. **(4 MARKS)**
 - b. Define and differentiate between functional and constraints in software engineering. **(4 MARKS)**
 - c. What is the significance of using a class diagram in software modeling? Illustrate with an example. **(5 MARKS)**
 - d. Discuss the different types of diagrams used in software modeling specifically in Structured Oriented Analysis. For each type, provide a brief explanation of its purpose and an example of when it might be used in a real-world scenario. **(12 MARKS)**
2.
 - a. Describe the steps involved in the requirement gathering process. Why is it crucial to accurately gather requirements in the early stages of software development? **(7 MARKS)**
 - b. What is a use case, and how does it differ from an entity relationship diagram? Use a scenario to illustrate your explanation. **(8 MARKS)**
 - c. Explain the concept of Object-Oriented Analysis and Design (OOAD). Discuss how OOAD principles can be applied to develop a robust software architecture. **(10 MARKS)**
3. You are tasked with designing a class diagram for a simple Library Management System. The system should include functionalities for managing books, members, and transactions (borrowing and returning books). Your class diagram should include at least three classes, with attributes, methods, and relationships (e.g., associations, generalizations) clearly defined. Provide explanations for your design choices and any assumptions you make. **(25 MARKS)**

4. Create an activity diagram to model the process of online shopping on an e-commerce platform. The process should include activities such as searching for products, adding items to the cart, checkout, payment processing, and order confirmation. Clearly label all activities, decision points, and control flows. Additionally, explain how this activity diagram can help in analyzing and improving the e-commerce platform's user experience. **(25 MARKS)**

-END OF QUESTION PAPER-